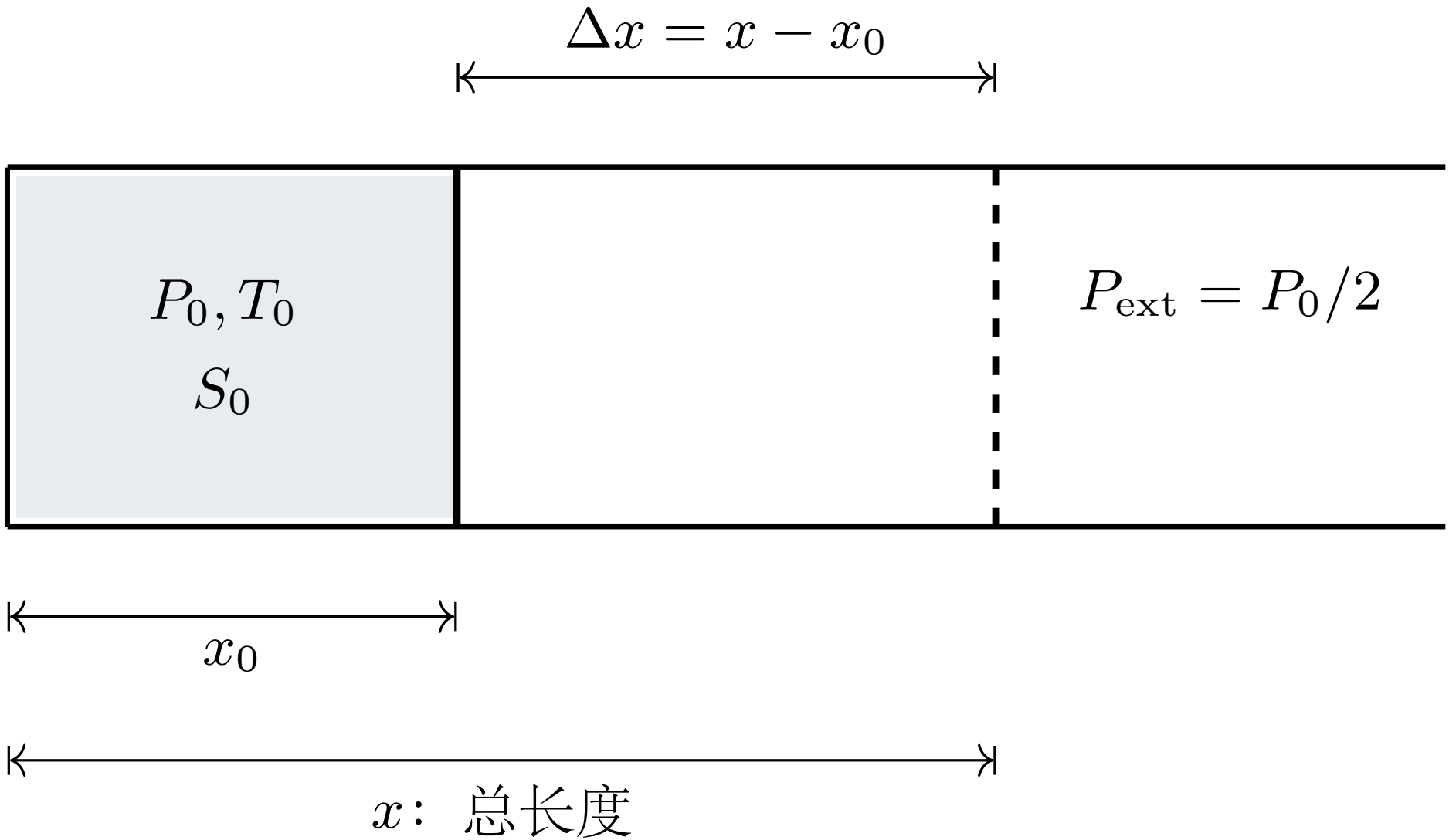
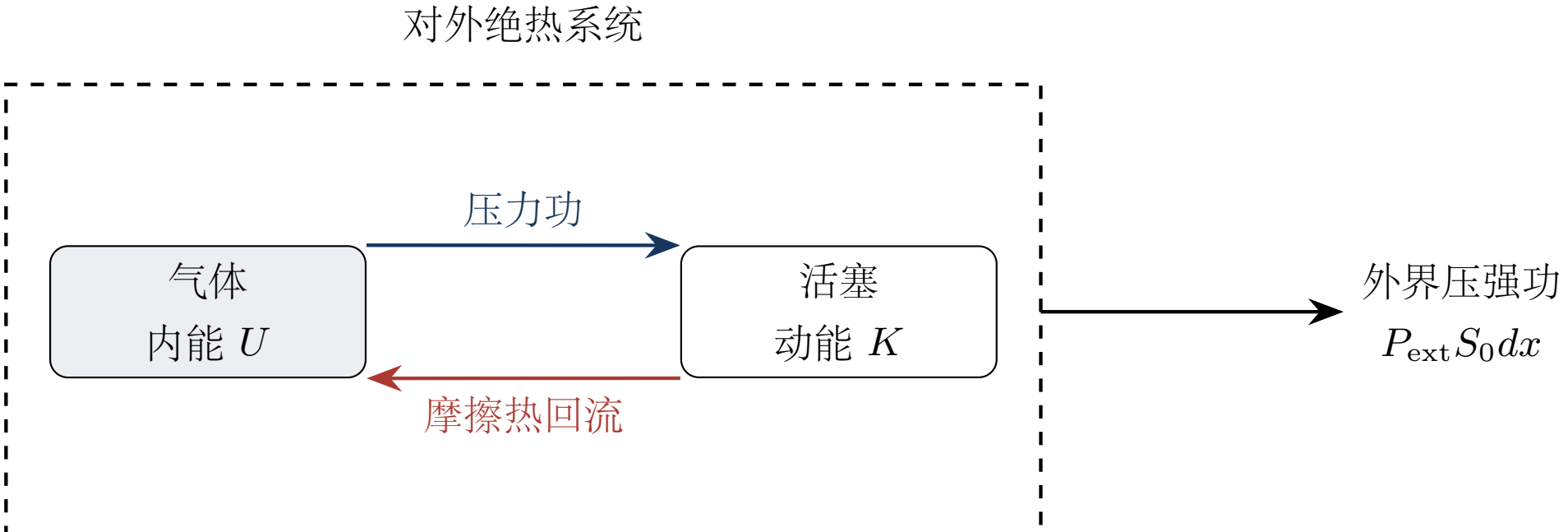
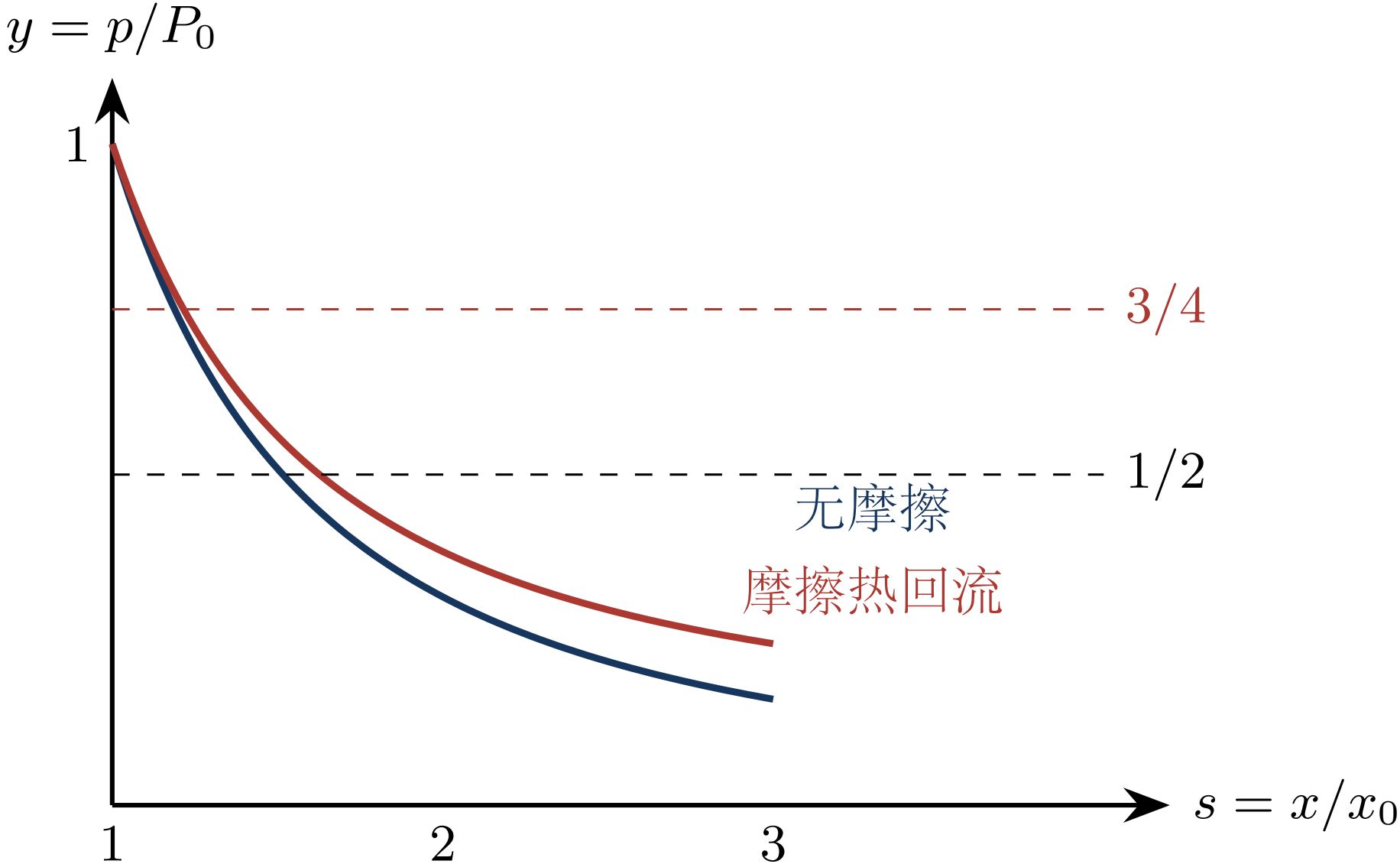
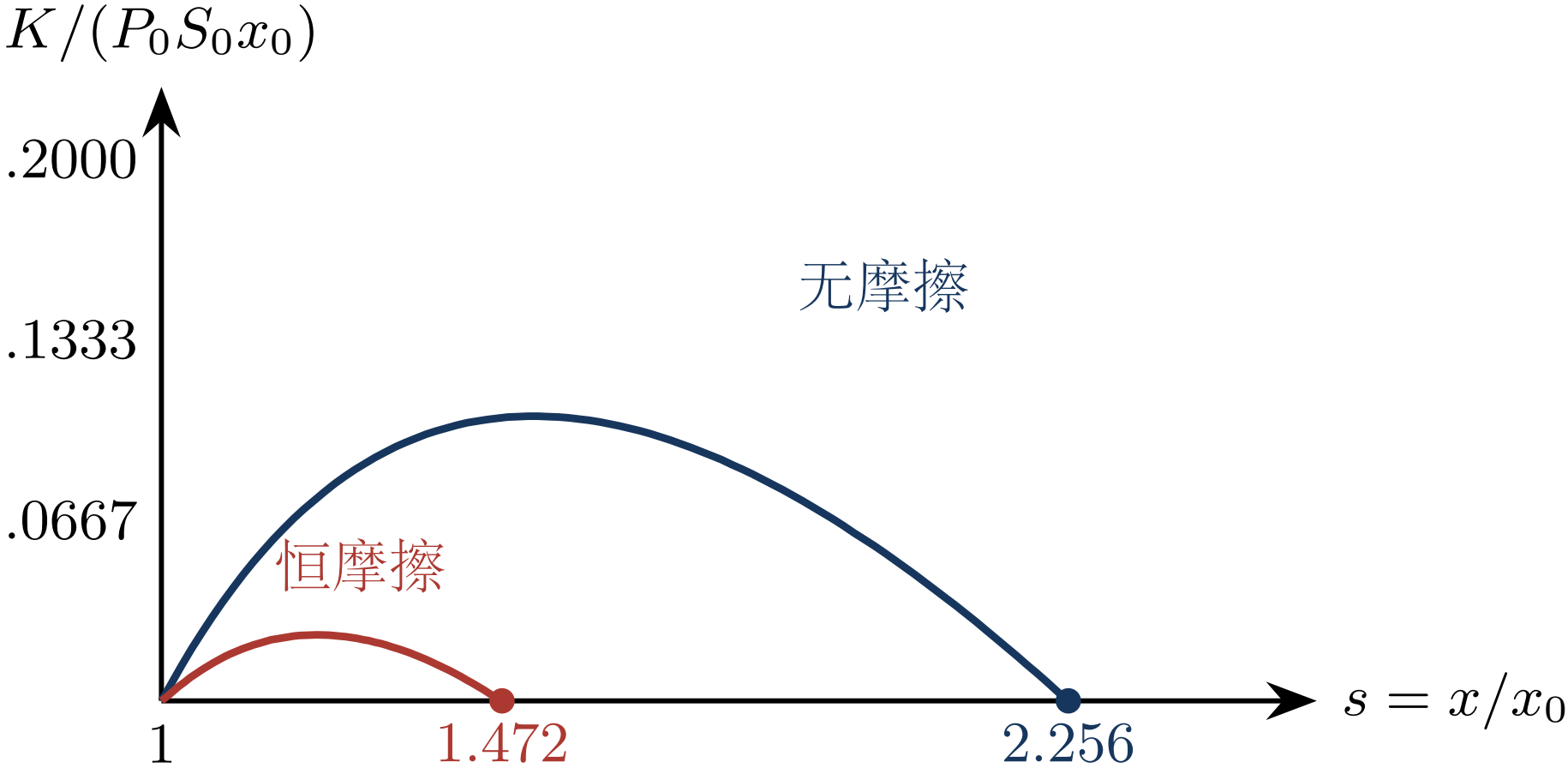

$\Delta x = x - x_0$
P_0, T_0
S_0
$P_{\text{ext}} = P_0/2$
x_0
x：总长度


对外绝热系统
压力功
气体
内能 U
活塞
动能 K
外界压强功
$P_{\text{ext}}S_0dx$
摩擦热回流


$y = p/P_0$
1
3/4
1/2
无摩擦
摩擦热回流
1
2
3
$s = x/x_0$


$K/(P_0S_0x_0)$
.2000
.1333
.0667
无摩擦
恒摩擦
1
1.472
2.256
$s = x/x_0$

长期平衡

$s_1 = 1.600$

$p/P_0 = 0.500$

无摩擦转向

$s_2 = 2.256$

$p/P_0 = 0.258$

恒摩擦零速

$s_3 = 1.472$

$p/P_0 = 0.573$

零速度条件与受力平衡条件分别判断